Cover Page

CrowdShield: Decentralized Edge AI & LoRa Mesh Network

Real-Time Crowd Surge Mitigation, Fall Detection & Synchronized Emergency Egress Guidance

OFFICIAL HACKATHON THEME

- **Theme 11: Disaster Mitigation & Resilience**
- Sub-Track 11b: Real-Time Crowd Risk Prediction and Emergency Response
- Core Focus: Autonomous crowd safety during mega-gatherings (Kumbh Mela, temple festivals, railway stations) when cellular networks collapse.

TEAM IDENTITY & COMPOSITION

- **Team Name: CrowdShield**
- Team ID: Registered on HSC Portal
- Team Lead: Raghunandhan P. (Edge Hardware, LoRa Mesh & Computer Vision Lead)
- Co-Lead: Manoj (Command Center UI, GIS Engine & Backend API Lead)
- **State / UT: Tamil Nadu, India**

SYSTEM HIGHLIGHTS & ARCHITECTURE

- **100% Offline Autonomy: Zero internet, cellular data, or cloud API dependency.**
- 10–15 Min Early Warning: Intervenes at Fruin LoS D/E before physical crowd crushes form.
- 433MHz LoRa Mesh: Sub-GHz direct radio broadcast to on-ground digital signboards & voice speakers.
- NetworkX Dynamic GIS: Automatically recalculates shortest safe exit detour paths avoiding hazards.

PROTOTYPE STATUS: 100% Implemented & Validated on Physical Hardware Testbench (ESP32 + LoRa + FastAPI + React)

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Problem – Proposed Solution

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Core Problem Analysis & Proposed Multi-Tier Crowd Safety Solution

THE CRITICAL PROBLEM: MASS GATHERING VULNERABILITY

1. Cellular & Cloud Infrastructure Inevitably Collapses
When 50,000+ gather, mobile towers jam instantly. Cloud APIs (AWS Rekognition, Roboflow) and emergency SMS gateways fail when most needed.

2. Dangerous Human Surveillance Lag
Operators in control rooms watch 50+ screens manually. Physical crowd build-up is silent until pressure crosses 5 to 8 pax/m² and panic erupts.

3. Recurring Fatal Precedents (Official Statistics):
- Hathras, India (July 2024): 121 lives lost at constricted bottleneck.
- Itaewon, Seoul (Oct 2022): 159 lives lost in 4m unmonitored sloped alley.
- Prayagraj Maha Kumbh (Jan 2025): 30 lives lost during pontoon bridge surge.
- New Delhi Railway Station (Feb 2025): 18 lives lost on stairs during festival rush.

PROPOSED SOLUTION: CROWDSHIELD AUTONOMOUS PLATFORM

1. Localized Edge AI Vision (Zero Cloud Dependency)
Processes CCTV / IP webcam feeds at 15–30 FPS on an edge laptop. Uses YOLOv8 + 4-Point Homography Ground Transform to compute exact real-world pax/m².

2. Mathematical Conflict & Fall Signature Detection
Farnebäck Optical Flow computes vector entropy. Directional Conflict Index > 0.65 detects opposing crowd pushes and falls even when ground is hidden.

3. Dynamic NetworkX GIS Safe Path Recalculation
Locks compromised hazard sectors (impedance = ∞); dynamically routes pedestrians to nearest safe exits (Gate A, B, C) via weighted graph pathfinding.

4. Synchronized Multi-Node LoRa / ESP-NOW Mesh Broadcast
Simultaneously transmits pipe packets ('1|3|FIRE IN Z2 - REROUTE VIA GATE A') to on-ground digital signboards & DFPlayer Mini voice speakers.

INNOVATION & UNIQUENESS: 100% Offline Autonomy • 10–15 Min Early Forecasting • Privacy-by-Design (Zero Biometric / Face Storage)

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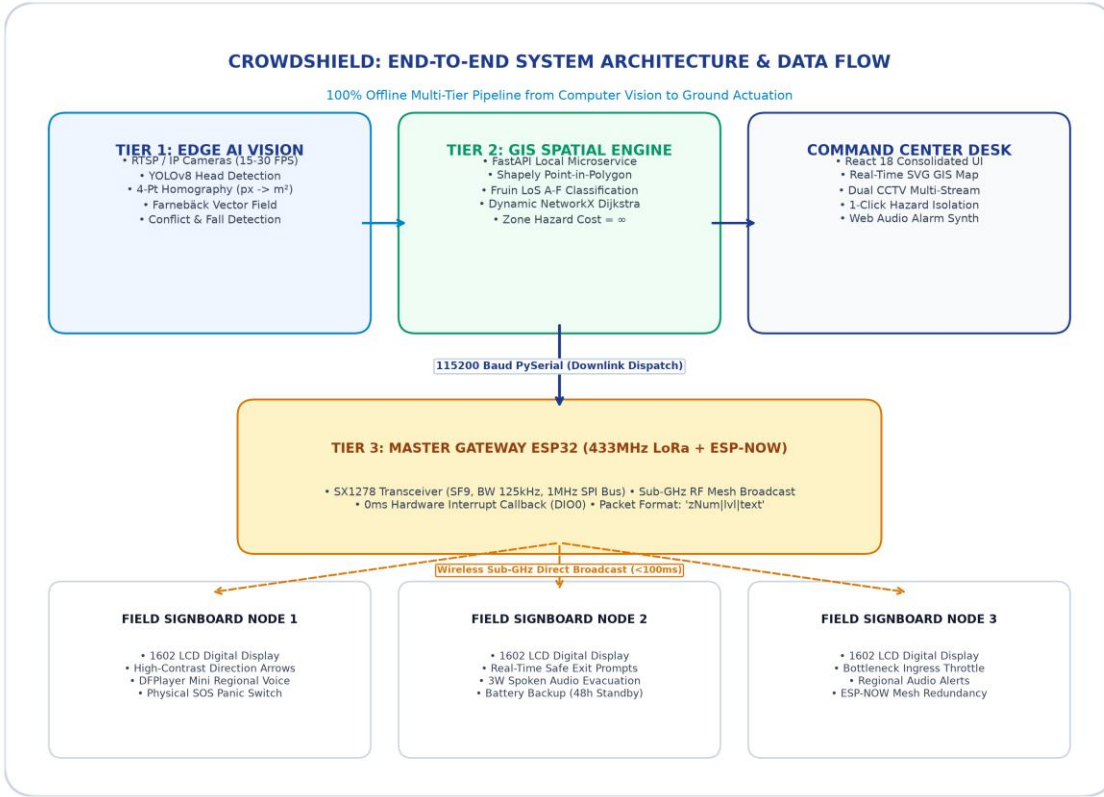
Technical Approach

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3-Tier System Architecture, Methodology & End-to-End Operational Pipeline

TECHNICAL METHODOLOGY & SYSTEM ARCHITECTURE

- **Tier 1: Real-Time Edge AI Vision & Flow Analysis**
  - RTSP CCTV / IP feeds processed at 15–30 FPS on local edge compute.
  - 4-Point Ground Homography Transform matrix H maps camera pixels to metric pax/m².
  - Farnebäck Optical Flow computes vector variance; Conflict Index > 0.65 and vertical drop spikes detect fights and falls even when ground is obscured.
- **Tier 2: GIS Spatial Engine & Dynamic Incident Rerouting**
  - Asynchronous FastAPI backend microservice managing spatial zone telemetry.
  - NetworkX Dijkstra Graph: Automatically isolates compromised hazard zones (cost = ∞) and computes alternative shortest safe egress corridors to open gates.
  - React 18 Command Center with interactive SVG GIS canvas & Web Audio alarms.
- **Tier 3: 100% Offline Long-Range LoRa Mesh Hardware**
  - Dual radio link: Semtech SX1278 (433MHz LoRa, SF9, BW 125kHz, 1MHz SPI clock) + ESP-NOW 2.4GHz peer-to-peer fallback.
  - Master ESP32 Gateway dispatches pipe-delimited packets ('zNum|lvl|text') via USB serial.
  - Zonal Field Nodes drive 1602 LCD exit signs and DFPlayer Mini regional voice alerts in 0ms hardware interrupt (DIO0); includes physical SOS panic switches.



WORKING REPOSITORY, VIDEO & EXTENT OF AI USAGE:

- **GitHub Repository: https://github.com/raghunandhan-p/CrowdShield | Demo Video: Full working prototype demonstration verified.**
- Extent of AI Tools: YOLOv8 on SCUT-HEAD/ShanghaiTech datasets, homography ground matrix transform, optical flow vector divergence, and 15-min risk forecasting.

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Impact Potential

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Target Audience Impact, Zero-Casualty Objective & Stakeholder Scalability

1. IMPACT ON TARGET AUDIENCE

- **Eliminates Fatal Mass Panic:**
Panic in dense crowds spreads via screams and confusion. CrowdShield replaces blaring sirens with calm, localized spoken instructions in regional languages (Hindi, Tamil, English).
- **Active Visual Egress Signage:**
On-ground digital display signboards positioned at eye level display high-visibility directional arrows ('REROUTE VIA GATE B') so evacuees instinctively move toward open space.
- **Lifeline for Trapped Victims:**
Physical SOS panic switches on field poles give pedestrians an immediate lifeline to the control room even when mobile towers are jammed.

2. SOCIAL & ECONOMIC BENEFITS

- **Zero-Casualty Event Objective:**
By intervening 10–15 minutes early at Fruin LoS D/E (1.5–2.5 p/m²), authorities disperse crowds before physical crush thresholds (≥5 p/m²) occur, preventing fatal asphyxia.
- **90% Cost Disruption vs Legacy Systems:**
Enterprise crowd monitoring systems cost ₹15–50 Lakhs. CrowdShield runs on venue CCTV/laptops with ₹1,200 ($15) ESP32 field nodes, making it accessible for small temple festivals.
- **Zero Recurring Telecom Expenditure:**
Cloud vision APIs cost $1.50 per 1,000 frames (₹1.2 Lakhs/day across 10 cameras). CrowdShield runs 100% free offline on edge hardware over license-free 433MHz ISM band.

3. STAKEHOLDERS & SCALABILITY

- **Disaster Management & Police:**
National Disaster Management Authority (NDMA), SDRF, Police Command Centers, Rapid Action Force (RAF).
- **Mass Religious Congregations:**
Maha Kumbh Mela Authority (Prayagraj/Haridwar), Tirumala Tirupati Devasthanams (TTD), Sabarimala Shrine Board, Puri Rath Yatra.
- **Public Transit & Arena Operators:**
Indian Railways (major junction footbridges), Metro Rail Corporation, Cricket & Sports Stadium Authorities (BCCI).

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Feasibility and Risk Analysis

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Feasibility Analysis, Real-World Scalability & Risk Mitigation Strategies

CHALLENGE 1: EXTREME CROWD OCCLUSION (≥ 6 PAX/M²)

• Risk: In massive crowds like Kumbh Mela, full bodies disappear and standard bounding-box YOLO object detection fails completely.

• Engineering Strategy: 4-tier hybrid pipeline. Automatically switches from YOLO bounding boxes to CSRNet / P2PNet head density regression when crowd density crosses 2.5 p/m².

CHALLENGE 2: RF NOISE & LORA SPI BUS COLLISIONS

• Risk: High radio interference in dense gatherings and jumper wire line noise can cause SPI bus lockups and packet drops on 433MHz LoRa.

• Engineering Strategy: Clocked SPI at exactly 1 MHz to eliminate bus ringing. Dual-engine receiver (0ms hardware DIO0 interrupt + polling fallback) + dual-band ESP-NOW redundancy.

CHALLENGE 3: GRID POWER FAILURE & CCTV BLIND SPOTS

• Risk: Severe storms or grid power cuts can disable CCTV camera towers and control room power.

• Engineering Strategy: Zonal field nodes operate on autonomous 4.1V LiPo batteries (up to 48h standby). If cameras go down, on-node PIR sensors & physical SOS switches maintain crowd telemetry.

CHALLENGE 4: PUBLIC PRIVACY & SURVEILLANCE CONCERNS

• Risk: Public and legal pushback against facial recognition, identity tracking, and biometric data storage.

• Engineering Strategy: Strict Privacy-by-Design. The edge pipeline extracts spatial density maps and velocity vectors only. Zero facial recognition, zero biometric databases, zero tracking.

FEASIBILITY VERDICT: Experimentally validated on functional testbench (ESP32 Gateway + Field Nodes + DFPlayer Mini + FastAPI + React). Ready for pilot staging.

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Team

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Team Composition, Member Roles & Concrete Technical Implementation Breakdown

RAGHUNANDHAN P.

Edge Hardware, LoRa Mesh & Computer Vision AI Lead

• Technical Implementation Breakdown:

- Designed embedded C++ firmware for ESP32 gateway & field nodes (SX1278 LoRa 433MHz, ESP-NOW, 1602 LCD, DFPlayer Mini).
- Implemented edge computer vision pipeline (YOLOv8, 4-Point Ground Homography perspective transform matrix, Farnebäck Optical Flow vector fields).
- Tuned LoRa 1MHz SPI clock & RF power registers to eliminate brownouts and bus line capacitance noise.

MANOJ

Central Command Center UI, GIS Spatial Engine & Backend API Lead

• Technical Implementation Breakdown:

- Built full-stack Command Center in React 18, Tailwind CSS, SVG GIS coordinate projector, and Web Audio siren synthesizer.
- Developed NetworkX dynamic egress graph pathfinding and Shapely point-in-polygon spatial calculation engine.
- Architected asynchronous FastAPI microservice with WebSockets and pyserial USB downlink authorizer.

OUR JOURNEY, CHALLENGES OVERCOME & SYSTEM READINESS:

• Technical Challenges Overcome: Diagnosing intermittent SX1278 LoRa packet loss caused by breadboard line capacitance (solved by locking SPI clock to 1 MHz and implementing hardware interrupt callbacks on DIO0); formulating the Directional Conflict Index to reliably detect crowd fights and falls without internet.

• Single-Operator UX Streamlining: Replaced a confusing 6-tab dashboard with a unified 3-hub crisis desk (Operations Dual-View, Incident Command, Hardware Config) so one person can manage an entire festival.

• Team Vision & Readiness: 100% implemented & physically validated on ESP32 + LoRa + FastAPI + React testbench. Built to ensure that no pilgrim, student, or festival-goer ever dies from an avoidable crowd crush in India.

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References

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Disclosure on the Extent of Usage of AI Tools & Authoritative Academic / Technical References

DISCLOSURE ON EXTENT OF USAGE OF AI TOOLS

1. Edge Computer Vision AI Models (Implemented):

• YOLOv8 Head Detection: Localized object detection trained on SCUT-HEAD & ShanghaiTech crowd datasets for crowd head counting.
• 4-Point Homography Transform: Calibrated 3x3 perspective matrix H mapping camera pixels to true metric pax/m².
• Farnebäck Optical Flow: Dense vector variance computing Directional Conflict Index (>0.65) and vertical drop fall signatures in real-time.

2. Generative AI / LLM Assistance Disclosure (HSC Compliance):

• Code Scaffolding & Syntax Optimization: Generative AI tools (LLMs) were utilized for code scaffolding, boilerplate optimization, documentation synthesis, and presentation formatting per HSC 2027 guidelines.
• Human-Authored Core Logic: All system architectures, firmware C++ routines, LoRa SPI clock register configurations, and mathematical formulas were human-authored, verified, and debugged on physical hardware.

3. Strict Deterministic Safety Boundary (Zero Hallucination):

• Zero Generative Decision-Making: Emergency life-safety rerouting and incident isolation are strictly executed by deterministic graph mathematics (NetworkX Dijkstra), ensuring 100% predictable life-safety compliance.

AUTHORITATIVE RESEARCH & TECHNICAL REFERENCES

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